Ques. What is Scope in Python?

- Every object in Python functions within a scope. A scope is a block of code where an object in Python remains relevant. Namespaces uniquely identify all the objects inside a program.
- Scope means **“WHERE can I use this variable?”**
- scope resolution in python follows the **LEGB rules**.
    - Local(L): Defined inside function/class
    - Enclosed(E): Defined inside enclosing functions(Nested function concept)
    - Global(G): Defined at the uppermost level
    - Built-in(B): Reserved names in Python builtin modules

1. **Global Scope/Global Variables:-** The Variable which can be **read from anywhere** in the program is known as a global scope. These variables can be accessed inside and outside the function.

```
x = 300 # global variable
def myfunc():
    print(x)

myfunc() # Outpur:- 300
print(x) # Outpur:- 300
```

2. **Local Variable**

```
def show():
    y = 5   # local variable
    print(y)

show()  # 5
print(y)   # ❌ Error - y cannot be accessed outside the function
```

3. **NonLocal or Enclosing Scope:-** Nonlocal Variable is the **variable that is defined in the nested function**. It means the variable can be neither in the local scope not in the global scope.

```
def func_outer():
    x = "local"
    def func_inner():
        nonlocal x
        x = "nonlocal"
        print("inner:", x)
    func_inner()
    print("outer:", x)
func_outer()

Output:-
inner: nonlocal
outer: nonlocal
```

3. **Built-in Scope:-** If a Variable is not defined in local, Enclosed or global scope, then python looks for it in the built-in scope. In the Following Example, 1 from math module pi is imported, and the value of pi is not defined in global, local and enclosed. Python then looks for the pi value in the built-in scope and prints the value. Hence the name which is already present in the built-in scope should not be used as an identifier.

```
# Built-in Scope
from math import pi
# pi = 'Not defined in global pi'
def func_outer():
    # pi = 'Not defined in outer pi'
    def inner():
        # pi = 'not defined in inner pi'
        print(pi)
    inner()
func_outer()

Output:- 3.141592
```

Same Variable Name, Different Scope

```
x = 10

def test():
    x = 5   # local variable
    print(x)

test()      # 5
print(x)    # 10
```